Provenance model methods and results outline

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1 Methods

1. First, we selected all the datasetIDstudy that had more than one provenance per species across the whole egret database ('provenanceVisualization.R'; nrow $n = 5959$. Second, we ran Victor's decision rules ('decisionRules.R') which yields 1449 rows of data. Third, we cleaned the dataset before fitting the model with forcing ('modelingCues.R'), yielding 1116 rows of data.
2. We fit the provenance model with and without forcing first, compared model estimates and moved on with the model without forcing with chilling and time as the main predictors. See the section 'comparing models' below for more details.
3. We compared the no-forcing model with the restricted (only the data with forcing information) versus the full dataset

$$p(\mu_i|\theta) = \begin{cases} 1 - logit^{-1}(\mu_i - k_1) & \text{if } \mu_i = 0 \\ [logit^{-1}(\mu - k_1) - logit^{-1}(\mu - k_2)] - \text{Beta\_proportion}(logit^{-1}(\mu), k_1) & \text{if } \mu_i \in]0, 1[\\ logit^{-1}(\mu_i - k_2) & \text{if } \mu_i = 1 \end{cases} \tag{1}$$

$$\mu_i = \alpha_{[sp]} + \alpha_{[prov]} + (\beta_{t[sp]} + \beta_{t[prov]}) \cdot \text{time}+ \tag{2}$$

$$(\beta_{f[sp]} + \beta_{f[prov]}) \cdot \text{forcing} + (\beta_{c[sp]} + \beta_{c[prov]}) \cdot \text{chilling} \tag{3}$$

$$\alpha_{[sp]} \sim \text{normal}(\mu_\alpha, \sigma_\alpha) \tag{4}$$

$$\alpha_{[prov]} \sim \text{normal}(0, \sigma_{[prov]}) \tag{5}$$

$$\tag{6}$$

2 Results

For the dataset that includes forcing (henceforth referred to as 'restricted'), we have 27 single species from 19 datasetID. For the dataset without forcing (full), we have 35 single species from 34 datasetID.

Comparing models

1. We fitted the model with forcing on the most restricted dataset, without and with provenance, which did not change the estimates (Figure 1)
2. Excluding forcing from the model did not influence the other parameters consistently across species or provenances
3. Forcing does not seem to explain the variation between the no-forcing and forcing models. Neither the maximum temperature (Figure 2), the number of unique germination treatments (Figure 3) or the temperature interval seem to explain the observed variation (Figure 4) between the restricted vs full dataset.
4. The estimates were unchanged when fitting the no-forcing model with the most restricted data ($n = 1116$) and the full dataset ($n = 1219$).

Results from selected model, i.e. the no-forcing model

1. Chilling increased germination for most species with huge variations. Four species probability to germinate decreased in response to chilling (Figure 6).

2. Time had a similar trend to forcing, with some more species increasing germination in response to time. Most of them didn't shift (Figure 7).

3. Germination varied more across species (σ = 2.943 (UI: 2.415, 3.552)) than across provenances (σ = 0.556 (UI: 0.45, 0.676) Figure 5)

4. We show global germination responses across all species as a function of chilling (Figure 8) and time (Figure 9).

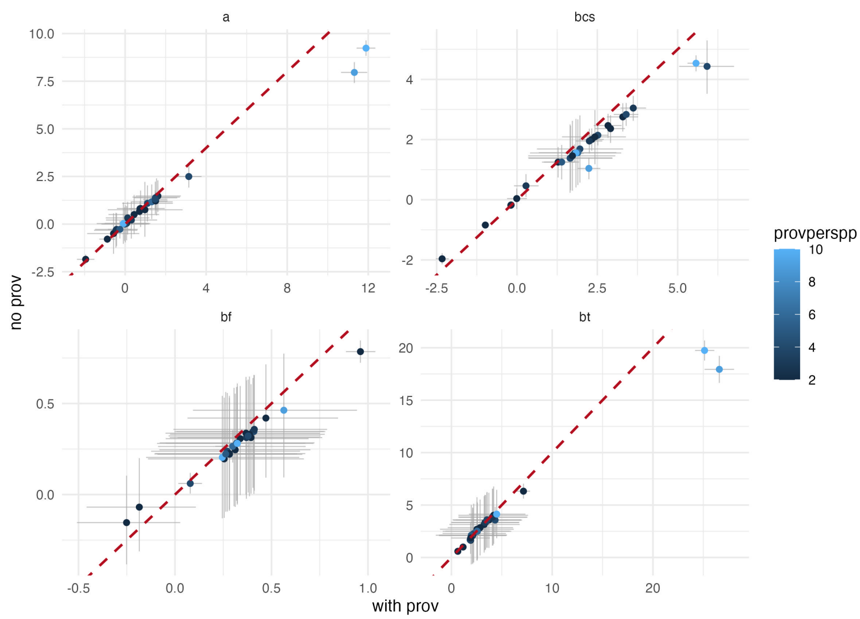

Figure 1: Comparing the no-provenance vs provenance models. Both of these models include the forcing parameters. We compare model parameters on a 1:1 line (red dashed line) where the y-axis represents the model that excludes provenance and the x-axis depicts the model that includes the provenance parameters. 'a' is the grand intercept for each species, 'bcs' is the chilling slope for each species, 'bf' is the forcing slope for each species and 'bt' is the time slope for each species.

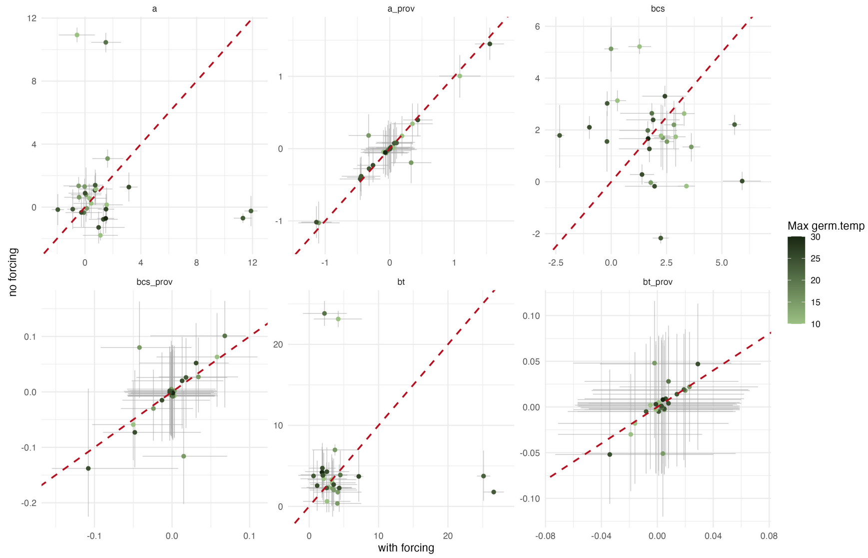

Figure 2: Comparing forcing vs no-forcing models color coded by the max temperature used in each study. We compare model parameters on a 1:1 line (red dashed line) where the x-axis represents the model that includes forcing and the y-axis depicts the model that excludes the forcing parameters. 'a' is the grand intercept for each species, 'a_prov' is the intercept for each provenance, 'bcs' is the chilling slope for each species, 'bcs_prov' is the chilling slope for each provenance, 'bt' is the time slope for each species and 'bt_prov' is the time slope for each provenance

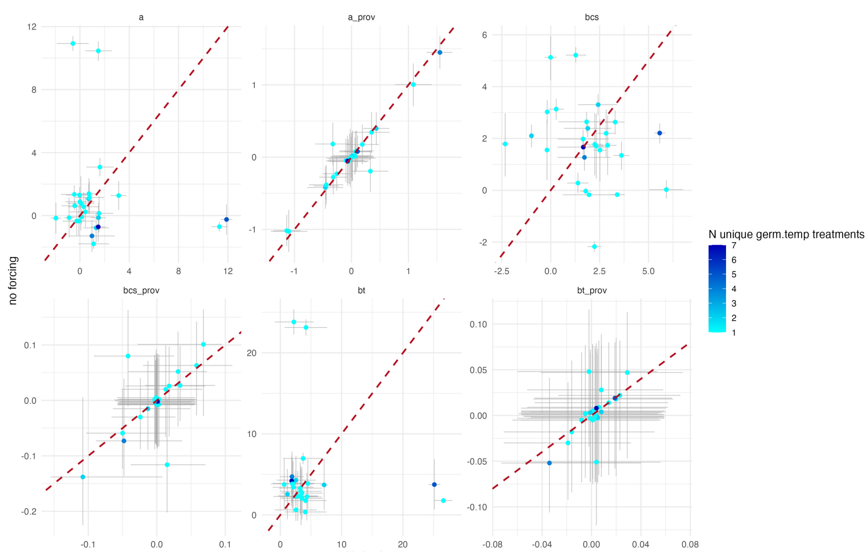

Figure 3: Comparing forcing vs no-forcing models color coded by the number of unique germination temperature treatments in each study. We compare model parameters on a 1:1 line (red dashed line) where the x-axis represents the model that includes forcing and the y-axis depicts the model that excludes the forcing parameters. 'a' is the grand intercept for each species, 'a_prov' is the intercept for each provenance, 'bcs' is the chilling slope for each species, 'bcs_prov' is the chilling slope for each provenance, 'bt' is the time slope for each species and 'bt_prov' is the time slope for each provenance

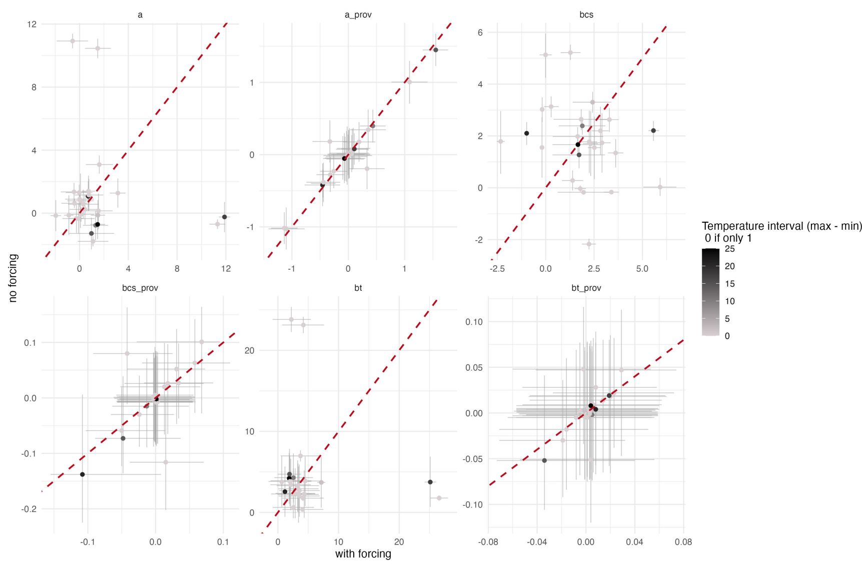

Figure 4: Comparing forcing vs no-forcing models color coded by the temperature interval in the study. We compare model parameters on a 1:1 line (red dashed line) where the x-axis represents the model that includes forcing and the y-axis depicts the model that excludes the forcing parameters. 'a' is the grand intercept for each species, 'a_prov' is the intercept for each provenance, 'bcs' is the chilling slope for each species, 'bcs_prov' is the chilling slope for each provenance, 'bt' is the time slope for each species and 'bt_prov' is the time slope for each provenance

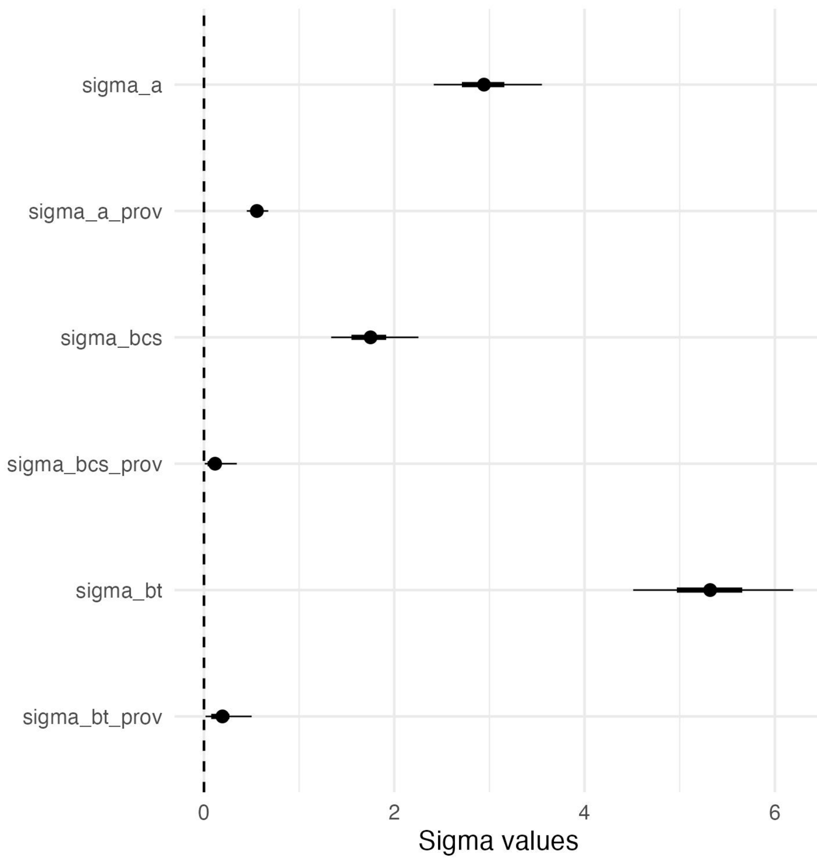

Figure 5: Results: sigma estimates for the no-forcing model. The dots represent the estimated mean of each provenance, with 50% uncertainty intervals (lines).

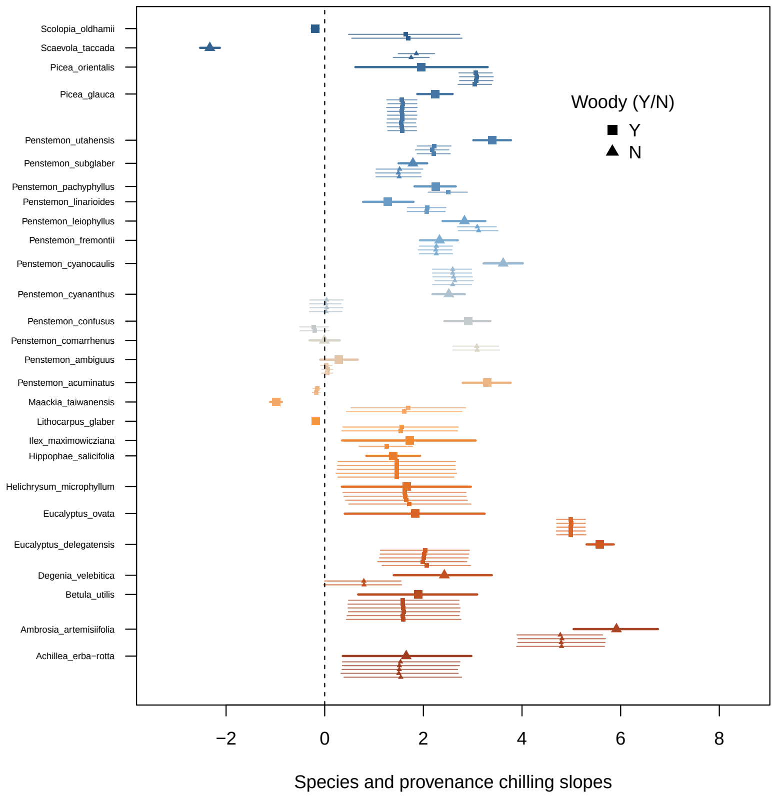

Figure 6: Results: chilling slope parameters for the no-forcing model.The dots represent the estimated mean of each provenance, with 50% uncertainty intervals (lines).

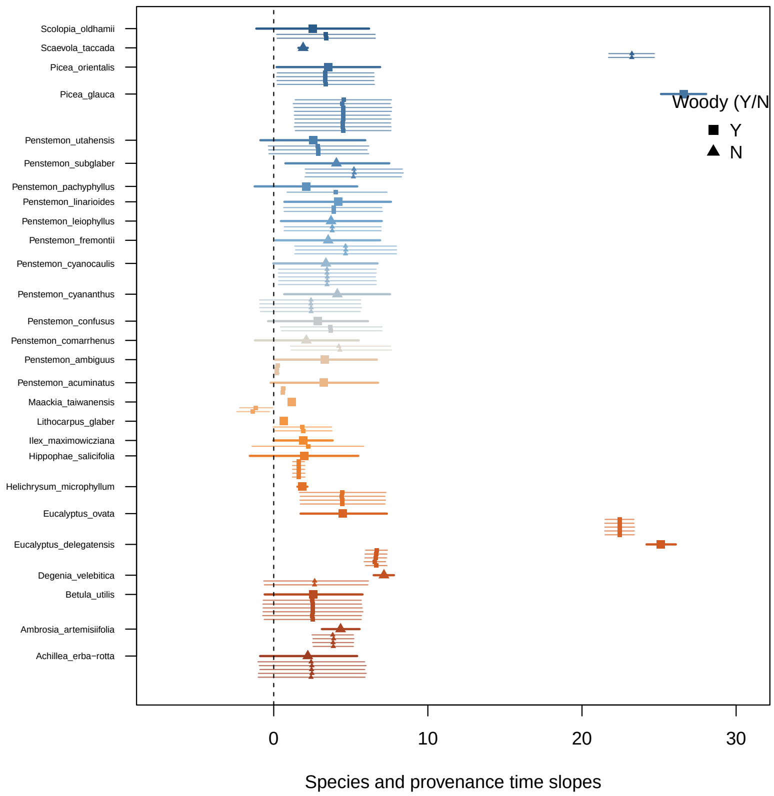

Figure 7: Results: time slope parameters for the no-forcing model. The dots represent the estimated mean of each provenance, with 50% uncertainty intervals (lines).

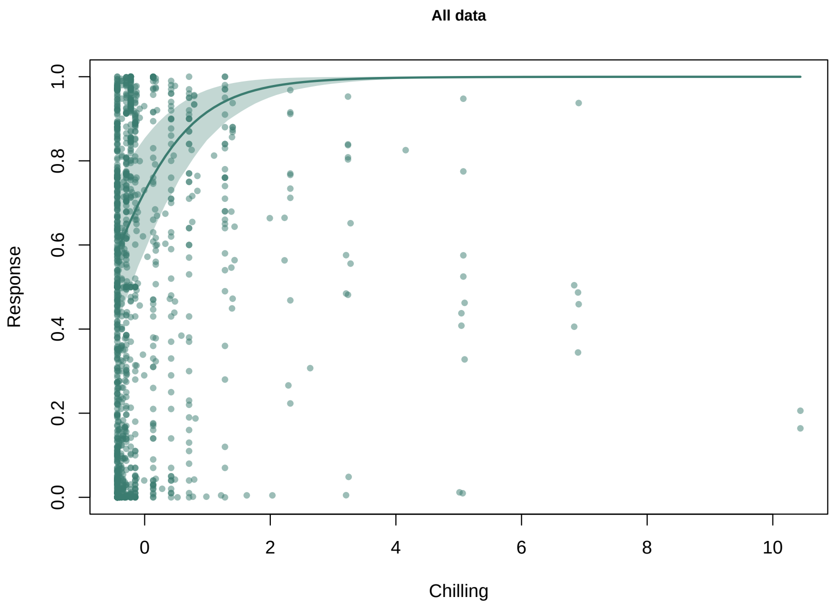

Figure 8: Results: general response curve across all species as a function of chilling (scaled) for the no-forcing model. We averaged the contribution of time and included it in the response curve.

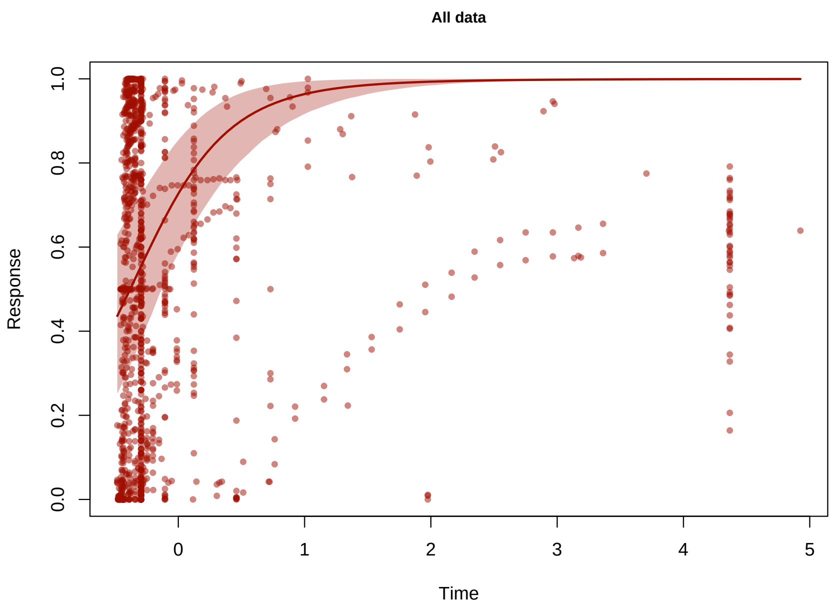

Figure 9: Results: general response curve across all species as a function of time (scaled) for the no-forcing model. We averaged the contribution of chilling and included it in the response curve.